RESTAURANT KITCHEN ORDER SCHEDULER

GROUP PROJECT REPORT

Submitted by

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Academic Year: 2026–2027

1. ABSTRACT

The **Restaurant Kitchen Order Scheduler** is a group project designed to arrange restaurant food orders efficiently. In a busy restaurant, many orders may arrive at the same time, and poor planning can cause delays.

The project uses **Priority Queue, Greedy Algorithm, and Job Sequencing** to decide which order should be prepared first. It considers preparation time, urgency, and promised delivery deadline.

The system demonstrates how Data Structures and Algorithms can solve a practical restaurant scheduling problem.

2. INTRODUCTION

Restaurants receive many food orders throughout the day. Some orders require less preparation time, while others may be urgent or have strict delivery deadlines.

This project creates a simple scheduler that accepts order details and generates a suitable kitchen preparation sequence.

3. PROBLEM STATEMENT

The restaurant needs to prepare several customer orders within their promised delivery times. Each order has an Order ID, food item, preparation time, urgency level, and deadline.

The main problem is to arrange orders so that important orders are handled efficiently and delays are reduced.

4. OBJECTIVES

- Arrange restaurant orders efficiently.
- Give priority to urgent orders.
- Consider delivery deadlines.
- Reduce total delay.
- Demonstrate Priority Queue and Heap concepts.
- Apply Greedy Algorithm and Job Sequencing.
- Display the final kitchen schedule clearly.
- Provide a simple and easy-to-use system.

5. GROUP PROJECT TEAM

Member	Suggested Responsibility
Member 1	Project planning, requirements and documentation
Member 2	Frontend design using HTML and CSS
Member 3	JavaScript/order-entry interface and validation
Member 4	C++ scheduling algorithms and Priority Queue
Member 5	Testing, results, presentation and final integration

These responsibilities can be changed according to the actual work completed by each member.

6. SCOPE OF THE PROJECT

- Add and store restaurant orders.
- Calculate order priority.
- Create a kitchen preparation schedule.
- Check delivery deadlines.
- Identify on-time and late orders.
- Calculate total delay.
- Display scheduling results.

7. TECHNOLOGIES USED

Component	Technology / Concept
Frontend	HTML, CSS, JavaScript
Backend / Algorithm Module	C++
Data Structure	Priority Queue / Heap
Algorithms	Greedy Algorithm, Job Sequencing, Sorting

8. SYSTEM REQUIREMENTS

Hardware:

- Computer or laptop
- Minimum 4 GB RAM
- Keyboard and mouse

Software:

- Windows / Linux / macOS
- C++ compiler
- Web browser
- Code editor such as VS Code

9. ORDER DATA

Field	Description
Order ID	Unique order number
Food Item	Name of food
Preparation Time	Time required to prepare food
Urgency	Importance of order
Deadline	Promised delivery time

Field	Description
Status	On Time or Late

Example:

Order	Food	Prep Time	Urgency	Deadline
O101	Pizza	15 min	8	12:30
O102	Burger	10 min	6	12:25
O103	Pasta	20 min	9	12:40

10. ALGORITHMS USED

Priority Queue: Orders are stored according to priority. Priority can consider earlier deadlines, higher urgency, and shorter preparation time.

Greedy Algorithm: The scheduler makes a suitable local choice at every step and updates kitchen time.

Job Sequencing: The system checks whether each scheduled order can finish before its promised deadline.

11. SYSTEM WORKING

Step 1: Enter order details

Step 2: Store order information

Step 3: Calculate priority

Step 4: Insert orders into the Priority Queue

Step 5: Select the best available order

Step 6: Schedule preparation

Step 7: Check the deadline

Step 8: Repeat until all orders are processed

Step 9: Display the final schedule

12. SYSTEM FLOW

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START → Enter Orders → Store Details → Calculate Priority → Priority Queue → Select Best Order → Schedule Preparation → Check Deadline → On Time / Late → Next Order → All Orders Done? → Display Result → END
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13. SAMPLE INPUT

Order	Preparation	Urgency	Deadline
O101	15 min	8	30 min
O102	10 min	6	25 min
O103	20 min	9	40 min

14. SAMPLE OUTPUT

Order	Start	Finish	Deadline	Status
O102	0 min	10 min	25 min	ON TIME
O101	10 min	25 min	30 min	ON TIME
O103	25 min	45 min	40 min	LATE

Summary: Total Orders: 3 | On-Time Orders: 2 | Late Orders: 1 | Total Delay: 5 minutes

15. USER INTERFACE

- Order Entry: Order ID, food name, preparation time, urgency and deadline.
- Priority Queue: Shows current order priority.
- Schedule: Shows order sequence and preparation times.
- Results: Shows total orders, on-time orders, late orders and total delay.

16. ADVANTAGES

- Simple order management.
- Better kitchen planning.
- Gives attention to urgent orders.
- Considers delivery deadlines.
- Reduces unnecessary delays.
- Demonstrates real-world use of algorithms.
- Easy to extend with additional features.

17. LIMITATIONS

- Basic version assumes one main scheduling queue.
- Chef availability is not considered.
- Different cooking stations are not considered.
- Preparation time is assumed to be known.
- Unexpected ingredient shortages are not included.
- Real-time traffic and customer location are not included.

18. FUTURE ENHANCEMENTS

- Multiple kitchen stations and chefs.
- Real-time order updates.
- Online food delivery integration.
- Customer notifications.
- Database storage.

- Staff login system.
- Mobile application.
- Machine-learning-based preparation-time prediction.

19. TIME COMPLEXITY

Operation	Time Complexity
Priority Queue insertion	O(log n)
Remove highest-priority order	O(log n)
Processing all orders	Approximately O(n log n)

Here, **n** represents the number of orders.

20. TESTING

Test Case	Expected Result
One order	Order is scheduled
Multiple orders	Orders are arranged by priority
Same deadline	Urgency is considered
Same urgency	Preparation time is considered
Very late order	Order is marked Late
Empty order list	No schedule is generated
Many orders	Priority Queue handles them efficiently

21. RESULT

The Restaurant Kitchen Order Scheduler successfully demonstrates the use of Data Structures and Algorithms for a real-world scheduling problem. The system accepts restaurant orders and generates a preparation sequence while checking deadlines and delays.

22. CONCLUSION

The **Restaurant Kitchen Order Scheduler** is a practical application of Data Structures and Algorithms. Priority Queue helps select important orders efficiently, while Greedy Scheduling and Job Sequencing help manage order priority and deadlines.

As a group project, the work combines interface design, algorithm implementation, testing, documentation, and presentation. The project provides a foundation that can be extended into a real-time restaurant kitchen management system.

23. REFERENCES

- Introduction to Algorithms – Cormen, Leiserson, Rivest and Stein.
- Data Structures and Algorithms – standard academic course materials.

- C++ Standard Library documentation.
- HTML, CSS and JavaScript documentation.
- Priority Queue and Heap concepts.
- Greedy Algorithm and Job Sequencing concepts.